Multi-platform distributed systems with Aggregate Computing in Kotlin: the case of proximity messaging

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Overview and Motivations

- Problem of centralized systems: single point of failure, low resilience, useless in the absence of a network.
- Real scenarios: emergencies, crowd messaging, widespread IoT

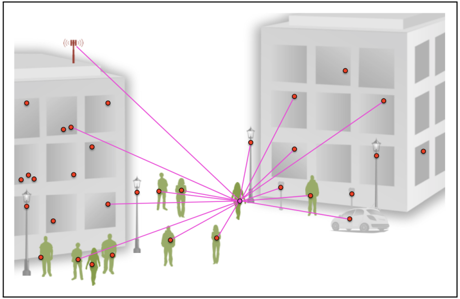

Figure: Example of a centralized system for proximity messaging.

Goal

Goal 1

Algorithm

Goal 2

Multi-platform mobile application with targets Android and iOS.

Aggregate Computing: The paradigm

Aggregate Computing

definition

Project Specifications

Multi-platform support

The project must be deployable on multiple platforms and targets.

- **Aggregate Computing framework**: Collektive
- **Programming language**: Kotlin
- **Multi-platform framework**: Kotlin Multiplatform
- **Simulator**: Alchemist Simulator

Algorithm (Gossip-Gradient)

Validation & CI

Mobile Application (Echo)

Conclusions & Future Development

The End